



Detection of avian pathogenic *Escherichia coli* (APEC) and antimicrobial resistance in layer parent stock in Bangladesh: Insights into phenotypic and genotypic profiles

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Highlights

- First study in Bangladesh on *Escherichia coli* in layer parent stock
- *Escherichia coli* was found in 82% of samples from live and dead birds.
- All isolates were multidrug-resistant; ampicillin resistance was 100%.
- APEC genes *fimC*, *iucD*, and *papC* were prevalent in *E. coli* isolates.
- Beta-lactam genes *bla*_{TEM}, *bla*_{CTX-M}, and *bla*_{SHV} were detected in the isolates.

Abstract

Avian pathogenic *Escherichia coli* (APEC) poses a major threat to poultry health and productivity, particularly in layer parent stock, resulting in substantial economic losses. This study aimed to determine the occurrence of APEC along with their virulence and resistance profile in the layer parent stock. A total of 89 samples, including cloacal swabs from 36 live birds and liver swabs from 53 dead birds, were collected. *E. coli* was isolated using culture, biochemical, and PCR methods. APEC-related virulence genes were detected via PCR, while antimicrobial susceptibility was evaluated using the disc diffusion method and PCR. The overall prevalence of *E. coli* was 82%, with similar detection rates in live (83.3%) and dead birds (81.1%). In PCR, APEC-associated gene *fimC* was detected in 95.9% of the isolates, *papC* in 36.9%, and *iucD* in 15.1%, with no significant difference between those from live and dead birds. All isolates were multidrug-resistant, where all *E. coli* isolates from both live and dead birds were resistant to ampicillin (100%), with high resistance to cefotaxime (live: 100% vs. dead: 93%), tetracycline (80% vs. 88.4%), nalidixic acid (100% vs. 62.8%), streptomycin (66% vs. 39.5%), norfloxacin (40% vs. 20.9%), and

chloramphenicol (20% vs. 25.6%). Moreover, genes conferring resistance to tetracyclines [*tetA* (live: 10/10 vs. dead: 10/10) and *tetB* (1/10 vs. 10/10)] and beta-lactams [*bla*_{TEM} (10/10 vs. 10/10), *bla*_{CTX-M} (1/10 vs. 4/10), and *bla*_{SHV} (2/10 vs. 0/10)] were detected in isolated *E. coli*. The results recommend periodic surveillance and policy interventions to mitigate health risks and economic losses.

Introduction

Avian pathogenic *Escherichia coli* (APEC) is a major bacterial pathogen responsible for significant economic losses in the poultry industry worldwide (Nawaz et al., 2024). It primarily affects chickens and other avian species, causing colibacillosis, which can manifest as localized infections, such as airsacculitis, pericarditis, and perihepatitis, or as systemic disease, including septicemia (Nawaz et al., 2024). Among commercial poultry, the layer parent stock represents a critical segment due to its role in sustaining egg production for future generations (Damaziak et al., 2021). Thus, the health and productivity of these flocks are of paramount importance, and APEC infections pose a direct threat to both animal welfare and production efficiency.

The *malB* gene, part of the *malB* operon in *E. coli*, encodes components of an ABC transporter system essential for the uptake and metabolism of maltose and maltodextrins (Boos and Shuman, 1998). Various virulence genes enhance the pathogenic potential of *E. coli* by facilitating host invasion, attachment, colonization, and evasion of immune responses. For example, *papC* (pyelonephritis-associated pili C) plays a role in producing adhesion structures that support bacterial attachment and infection, while *fimC* (Type 1 fimbriae C) contributes to epithelial adherence and colonization. The *iucD* gene, involved in iron acquisition, helps distinguish APEC from non-pathogenic strains. The overall virulence of an *E. coli* isolate is influenced by both the presence and combination of these specific virulence determinants (Ievy et al., 2020; Islam et al., 2021; Mohamed et al., 2018; Subedi et al., 2018; Zakariazadeh et al., 2019).

In layer parent stock, APEC infections can lead to poor reproductive performance, increased morbidity, mortality, and reduced egg quality, ultimately affecting hatchability and flock longevity (Watts and Wigley, 2024). Understanding the prevalence and characteristics of APEC in this group of poultry is crucial for developing effective preventive measures. However, the challenge is compounded by the pathogen's ability to acquire and disseminate antimicrobial resistance (AMR), particularly in environments where antimicrobial agents are frequently used, both therapeutically and prophylactically. AMR in APEC strains complicates the management of infections, limiting treatment options and raising concerns about the potential for zoonotic transmission to humans (Jerzsele et al., 2025).

In recent years, the issue of AMR has gained global attention due to the increasing emergence of multidrug-resistant (MDR) bacteria in both human and veterinary medicine (Naghavi et al., 2024). In poultry production systems, the indiscriminate or inappropriate use of antibiotics has been a major contributing factor to the development of resistant strains (Abreu et al., 2023). The emergence of AMR in APEC represents a critical intersection of animal health, food safety, and public health. Resistant APEC strains circulating in poultry populations can not only impair treatment efficacy but also potentially serve as reservoirs of resistance genes, with the potential to spread to other bacterial species and even to humans via the food chain (Naghavi et al., 2024).

Layer parent stocks, due to their prolonged production cycle and high economic value, are often subjected to antimicrobial treatments aimed at preventing or controlling bacterial infections (Gržinić et al., 2023). However, this population is also at a heightened risk of harboring resistant bacteria, including APEC. To better manage colibacillosis and minimize the risks associated with AMR, it is essential to identify and characterize the specific APEC strains circulating within layer parent stock and assess their AMR profiles. Therefore, this study aimed to fill a critical gap in literature by focusing on the detection and characterization of APEC strains in layer parent stock, along with a comprehensive analysis of their antimicrobial resistance profiles using phenotypic and genotypic assays.

Section snippets

Ethical statement

All procedures involving animals were conducted in accordance with the ethical standards and guidelines approved by the Bangladesh Agricultural University (BAU) Animal Welfare and Experimentation Ethics Committee (AWEEC). The research protocol was reviewed and approved under approval number AWEEC/BAU/2024(2)/20(a), dated 01-12-2024. ...

Sampling information and sample processing

This study was carried out in the Bhaluka Upazila (24.3789° N, 90.19373° E) of Mymensingh District in Bangladesh. The availability and prominence of layer breeder ...

Occurrence of *E. coli* in layer parent stock

In PCR, based on the presence of the *malB* gene (Fig. 2A), the overall occurrence of *E. coli* was 82% (95% CI, 72.77–88.62%), where the rate was recorded higher ($P>0.05$) in cloacal swabs of live birds (83.33%, 95% CI, 68.10–92.13%) than in liver swabs of dead birds at 81.13% (95% CI, 68.64–89.41%). ...

Occurrence of virulence genes of *E. coli*

PCR revealed the presence of APEC-related virulence genes in *E. coli* isolates (Fig. 2B, C, and D). The APEC-associated virulence genes *fimC* and *iucD* were detected more frequently in *E. coli* ...

Prevalence of *E. coli* in layer parent stock

The present study comprehensively assessed the occurrence, virulence, and antimicrobial resistance profiles of *E. coli* isolated from live and dead layer parent stock in Bangladesh. The overall occurrence of *E. coli* in the studied layer parent stock was 82%, with slightly higher rates in cloacal swabs from live birds (83.33%) compared to liver swabs from dead birds (81.13%). This high prevalence is consistent with a previous study, which reported *E. coli* as a common inhabitant of the avian ...

Conclusions

This study highlights the high prevalence of *E. coli*, APEC-associated virulence genes, and MDR in layer parent stock in Bangladesh. The findings underscore the need for enhanced biosecurity, antimicrobial stewardship, and disease prevention measures in poultry farming to protect both animal and human health. Addressing the challenges posed by AMR in poultry requires a coordinated, multisectoral approach that integrates the principles of One Health. Moreover, as a further critical step in ...

CRediT authorship contribution statement

Md. Nowshad Atiq: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Md. Saiful Islam:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Md. Ashek Ullah:** Investigation. **Nirab Chakroborty:** Investigation. **Raihana Islam:** Investigation. **Md. Liton Rana:** Investigation. **Farhana Binte Ferdous:** Investigation. **Md. Abdus Sobur:** Investigation. **Mohammad ...**

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Declaration of competing interest

The authors declare no conflict of interest. The funders had no role in the design of the study, the collection, analysis, or interpretation of data, the writing of the manuscript, or the decision to publish the results. ...

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